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Harmonic, the Robinhood CEO's AI math startup, launches an AI chatbot app

Maxwell Zeff — 3:27 PM PDT · July 28, 2025



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Eventually, Harmonic says it also plans to release an API to let enterprises access Aristotle, as well as a web app for consumers.

Harmonic says Aristotle achieved gold medal performance on the 2025 International Math Olympiad (IMO) through a formal test (meaning the problems were translated into a machine-readable

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format). Google and OpenAI also developed AI models that achieved gold medal performance on this year's IMO, but through informal tests taken in natural language.

Harmonic says it's not releasing other benchmarks for Aristotle at this time.

The beta launch of Aristotle comes just a few weeks since Harmonic raised \$100 million in a Series B



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Several leading tech companies are focused on training their AI models to solve math problems. AI that can do math is valuable in its own right, but math is also considered a uniquely verifiable domain that demands core reasoning skills. Systems that develop these capabilities may prove useful in other domains as well.

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Achim says that Harmonic achieves its hyper-accurate solutions by having Aristotle produce responses in the open source programming language Lean. Before Aristotle gives an answer to users, he says the model double-checks that the solution is correct through an algorithmic process that doesn't involve AI. Harmonic's CEO notes that similar technology is used to verify outputs in high-stakes fields such as medical devices and aviation.

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AI



DeepMind claims its newest AI tool is a whiz at math and science problems



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Because it can self-evaluate, the system can only work with certain types of problems — specifically those in fields like computer science and system optimization. In another major limitation, AlphaEvolve can only describe solutions as algorithms, making it a poor fit for problems that aren't numerical.

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DeepMind, however, is making the same case that many AI labs do for their systems: that AlphaEvolve can save time while freeing up experts to focus on other, more important work.

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OpenAI's new reasoning
AI models hallucinate
more

OpenAI's [recently launched o3 and o4-mini AI models](#) are state-of-the-art in many respects.

However, the new models still hallucinate, or make

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understand why hallucinations are getting worse as it scales up reasoning models. O3 and o4-mini perform better in some areas, including tasks related to coding and math. But because they “make more claims overall,” they’re often led to make “more accurate claims as well as more inaccurate/hallucinated claims,” per the report.

OpenAI found that o3 hallucinated in response to 33% of questions on PersonQA, the company's in-house benchmark for measuring the accuracy of a model's knowledge about people. That's roughly double the hallucination rate of OpenAI's previous reasoning models, o1 and o3-mini, which scored 16% and 14.8%, respectively. O4-mini did even worse on PersonQA — hallucinating 48% of the time.

Third-party research tends to observe a number of some to



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“Our hypothesis is that the kind of reinforcement learning used for o-series models may amplify issues that are usually mitigated (but not fully

erased) by standard post-training pipelines,” said Neil Chowdhury, a Transluce researcher and former OpenAI employee, in an email to TechCrunch.

Sarah Schwettmann, co-founder of Transluce, added that o3’s hallucination rate may make it less useful than it otherwise would be.

Kian Katanforoosh, a Stanford adjunct professor and CEO of the upskilling startup Workera, told



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If scaling up reasoning models indeed continues to worsen hallucinations, it’ll make the hunt for a solution all the more urgent.

“Addressing hallucinations across all our models is an ongoing area of research, and we’re continually working to improve their accuracy and reliability,”

said OpenAI spokesperson Niko Felix in an email to TechCrunch.

In the last year, the broader AI industry has pivoted to focus on reasoning models after [techniques to improve traditional AI models started showing diminishing returns](#). Reasoning improves model performance on a variety of tasks without requiring massive amounts of computing and data during training. Yet it seems reasoning also may lead to more ha

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